Wrocław University
of Science and Technology

Optimization Methods
Project Group 1

Task 0: Implementation and Study of Evolutionary Algorithms Compared to Non-evolutionary Methods

The Permutation Flowshop Scheduling Problem
— Project's Report —

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Contents

1 The Permutation Flowshop Scheduling Problem **2**
1.1 General overview . . . 2
1.2 Specifics . . . 2
1.3 Objectives . . . 2
1.4 Constraints . . . 2
1.5 Chosen problem representation . . . 3
1.6 Solved example . . . 3

2 Theoretical review **4**
2.1 Genetic Algorithms . . . 4
2.2 ds . . . 4

3 Implementation **4**
3.1 Main structure . . . 4
3.2 Random Search . . . 4
3.3 Greedy . . . 4
3.4 EA . . . 4
3.5 Simulated Annealing . . . 4

4 Results and analysis **4**
4.1 Description of the different cases . . . 5
4.2 Basic instances . . . 6
4.2.1 Instance 1: 20 jobs and 5 machines . . . 6
4.2.2 Instance 2 : 20 jobs and 10 machines . . . 6
4.2.3 Instance 3 : 20 jobs and 20 machines . . . 6
4.3 Medium instances . . . 7
4.3.1 Instance 4 : 100 jobs and 10 machines . . . 7
4.3.2 Instance 5 : 100 jobs and 20 machines . . . 7
4.4 Hard instance : 500 jobs and 20 machines . . . 7
4.5 notepadd graphics . . . 7
4.6 Evolution . . . 7

Conclusion **7**

Appendixes **8**
List of Tables . . . 8
List of Figures . . . 8

Introduction

This documents aims to present the main results on the *Permutation Flowshop Scheduling Problem* (PFSP).

Context

Presentation of the plan

1 The Permutation Flowshop Scheduling Problem

1.1 General overview

The PFSP is defined by a set of M machines, J jobs and a matrix $P \in \mathcal{M}_{M,J}(\mathbb{R})$, where $p_{m,j} = p(m,j)$ represents the processing times of job j on machine m with $j \in [\![1; J]\!]$ and $m \in [\![1; M]\!]$.

The constraints are that each job must be processed on each machine. Moreover, the order of machines on which jobs are processed is predefined, which mean that we cannot start processing a job on a machine until the job is done on the previous machine, and the order is the same for all jobs. Finally, each machine can process only one job at a time.

We aim to find a job sequence $x = \langle x_i, \dots, x_n \rangle$ to be processed on each machine which minimizes the total flow time. We define the total flow time as the sum of the completion times of each job on the last machine. Hence, the objective function f that compute the total flow is defined as:

$$f(x) = \sum_{j=1}^{J} c(x_j, M), \tag{1}$$

where $c(x_j, m)$ is the completion time of the x_j-th job on the $m-$th machine. This quantity can be calculated with

$$c(x_j, m) = \begin{cases} p(x_j, m) & \text{, if } j = 1 \text{ and } m = 1 \\ c(x_j, m-1) + p(x_j, m) & \text{, if } j = 1 \text{ and } m > 1 \\ c(x_{j-1}, m) + p(x_j, m) & \text{, if } j > 1 \text{ and } m = 1 \\ \max\{c(x_{j-1}, m), c(x_j, m-1)\} + p(x_j, m) & \text{, if } j > 1 \text{ and } m > 1 \end{cases}, \tag{2}$$

where $p(x_j, m)$ is the processing time on job x_j on machine m.

1.2 Specifics

the goal sould target a specific area of improvement or answer a specific need

1.3 Objectives

1.4 Constraints

1.5 Chosen problem representation

1.6 Solved example

Let's consider the following problem with 4 jobs and 5 machines and the given computation times matrix (3).

$$P = \begin{pmatrix} 30 & 29 & 19 & 50 \\ 39 & 69 & 6 & 73 \\ 3 & 86 & 58 & 60 \\ 72 & 74 & 51 & 15 \\ 24 & 47 & 22 & 90 \end{pmatrix} \tag{3}$$

For this problem, we know the optimal solution, which is the sequence $x^* = \langle 1, 3, 4, 2 \rangle$.

By using equation (1), we can compute the total flow time:

$$f(x^*) = c(1,5) + c(3,5) + c(4,5) + c(2,5)$$

Now, we use (2) to evaluate each cost. We present the calculations in the reverse order necessary to understand the dependencies between the relationships

- **Calculation of** $c(1,5)$
  - We can in fact prove by recurrence that, for all $n \in \mathbb{N}$, we have $c(1,n) = \sum_{j=1}^{n} p(1,j)$.
  - For $n = 5$, we have indeed:

$$\begin{aligned} c(1,5) &= c(1,4) + p(1,5) \\ &= c(1,3) + p(1,4) + p(1,5) \\ &= \cdots \\ &= p(1,1) + p(1,2) + p(1,3) + p(1,4) + p(1,5) \\ &= 30 + 39 + 3 + 72 + 24 \\ \boxed{c(1,5) = 168} \end{aligned}$$

- **Calculation of** $c(3,5)$
  - $c(1,2) = p(1,1) + p(1,2) = 40 + 39 = 69$
  - $c(3,1) = c(1,1) + p(3,1) = p(1,1) + p(3,1) = 30 + 19 = 49$
  - $c(3,2) = \max\{c(1,2), c(3,1)\} + p(3,2) = c(1,2) + p(3,2) = 69 + 6 = 75$
  - $c(3,3) = \max\{c(1,3), c(3,2)\} + p(3,3) = c(3,2) + p(3,3) = 75 + 58 = 133$
  - $c(3,4) = \max\{c(1,4), c(3,3)\} + p(3,4) = c(1,4) + p(3,4) = 144 + 51 = 195$

- $c(3,5) = \max\{c(1,5), c(3,4)\} + p(3,5) = c(3,4) + p(3,5) = 195 + 22 = 217$
- $\boxed{c(3,5) = 217}$

- **Calculation of** $c(4,5)$
  - $c(4,1) = c(3,1) + p(4,1) = 49 + 50 = 99$
  - $c(4,2) = \max\{c(3,2), c(4,1)\} + p(4,2) = c(4,1) + p(4,2) = 99 + 73 = 172$
  - $c(4,3) = \max\{c(3,3), c(4,2)\} + p(4,3) = c(4,2) + p(4,3) = 172 + 60 = 232$
  - $c(4,4) = \max\{c(3,4), c(4,3)\} + p(4,4) = c(4,3) + p(4,4) = 232 + 15 = 247$
  - $\boxed{c(4,5) = 247}$

- **Calculation of** $c(2,5)$
  - $c(2,1) = c(4,1) + p(2,1) = 99 + 29 = 128$
  - $c(2,2) = \max\{c(4,2), c(2,1)\} + p(2,2) = c(4,2) + p(2,2) = 172 + 69 = 241$
  - $c(2,3) = \max\{c(4,3), c(2,2)\} + p(2,3) = c(2,2) + p(2,3) = 241 + 86 = 327$
  - $c(2,4) = \max\{c(4,4), c(2,3)\} + p(2,4) = c(2,3) + p(2,4) = 327 + 74 = 401$
  - $c(2,5) = \max\{c(4,5), c(2,4)\} + p(2,5) = c(2,4) + p(2,5) = 401 + 47 = 448$
  - $\boxed{c(2,5) = 448}$

- Finally, we get $\boxed{f(x^*) = 168 + 217 + 337 + 448 = 1170}$

2 Theoretical review

2.1 Genetic Algorithms

2.2 ds

3 Implementation

3.1 Main structure

3.2 Random Search

3.3 Greedy

3.4 EA

3.5 Simulated Annealing

4 Results and analysis

4.1 Description of the different cases

Case n°	**Population size**	**Generations**	P_x	P_m	Tour	Mutation	Crossover	**Run per algo**
1	100	100	0.7	0.1	5	Swap	OX	10
2	1000	100	0.7	0.1	5	Swap	OX	10
3	10000	100	0.7	0.1	5	Swap	OX	10

Table 1: **Description of the parameters of the cases using to determine the best population size.**

Case n°	**Population size**	**Generations**	P_x	P_m	Tour	Mutation	Crossover	**Run per algo**
4	a^*	100	0.7	0.1	5	Swap	OX	10
5	a^*	1000	0.7	0.1	5	Swap	OX	10
6	a^*	10000	0.7	0.1	5	Swap	OX	10

Table 2: **Description of the parameters of the cases using to determine the best number of generations.**

Case n°	**Population size**	**Generations**	P_x	P_m	Tour	Mutation	Crossover	**Run per algo**
7	a^*	b^*	0.6	0.1	5	Swap	OX	10
8	a^*	b^*	0.7	0.1	5	Swap	OX	10
9	a^*	b^*	0.8	0.1	5	Swap	OX	10

Table 3: **Description of the parameters of the cases using to determine the best crossover probability.**

Case n°	**Population size**	**Generations**	P_x	P_m	Tour	Mutation	Crossover	**Run per algo**
10	a^*	b^*	c^*	0.1	5	Swap	OX	10
11	a^*	b^*	c^*	0.2	5	Swap	OX	10
12	a^*	b^*	c^*	0.3	5	Swap	OX	10

Table 4: **Description of the parameters of the cases using to determine the best mutation probability.**

Case n°	Population size	Generations	P_x	P_m	Tour	Mutation	Crossover	Run per algo
13	a^*	b^*	c^*	d^*	5	Swap	OX	10
14	a^*	b^*	c^*	d^*	10	Swap	OX	10
15	a^*	b^*	c^*	d^*	15	Swap	OX	10

Table 5: **Description of the parameters of the cases using to determine the best tour size.**

Case n°	Population size	Generations	P_x	P_m	Tour	Mutation	Crossover	Run per algo
16	a^*	b^*	c^*	d^*	e^*	Swap	OX	10
17	a^*	b^*	c^*	d^*	e^*	Inversion	OX	10

Table 6: **Description of the parameters of the cases using to determine the best mutation type.**

Case n°	Population size	Generations	P_x	P_m	Tour	Mutation	Crossover	Run per algo
18	a^*	b^*	c^*	d^*	e^*	f^*	OX	10
19	a^*	b^*	c^*	d^*	e^*	f^*	PMX	10

Table 7: **Description of the parameters of the cases using to determine the best crossover type.**

4.2 Basic instances

4.2.1 Instance 1: 20 jobs and 5 machines

	Random Alg. [10k]				Greedy Alg.	Evol. Alg. [10x]				TA or SA [10x]			
Case n°	**best***	**worst**	**avg**	**std**	**best***	**best***	**worst**	**avg**	**std**	**best***	**worst**	**avg**	**std**
1	15124	15745	15407.6	168.6	16384	14127	14406	14226.1	93.0	**14269**	14604	14424.6	101.9
Time	12.59 s				0.0084 s	15.42 s				12.32 s			
2	15124	15745	15407.6	168.6	16384	14127	14406	14226.1	93.0	**14269**	14604	14424.6	101.9
Time	12.59 s				0.0084 s	15.42 s				12.32 s			

Table 8: Determine the best population size

4.2.2 Instance 2 : 20 jobs and 10 machines

4.2.3 Instance 3 : 20 jobs and 20 machines

4.3 Medium instances

4.3.1 Instance 4 : 100 jobs and 10 machines

4.3.2 Instance 5 : 100 jobs and 20 machines

4.4 Hard instance : 500 jobs and 20 machines

4.5 notepadd graphics

4.6 Evolution

Conclusion

Appendixes

List of Tables

1 **Description of the parameters of the cases using to determine the best population size.** . . . 5
2 **Description of the parameters of the cases using to determine the best number of generations.** . . . 5
3 **Description of the parameters of the cases using to determine the best crossover probability.** . . . 5
4 **Description of the parameters of the cases using to determine the best mutation probability.** . . . 5
5 **Description of the parameters of the cases using to determine the best tour size.** . . . 6
6 **Description of the parameters of the cases using to determine the best mutation type.** . . . 6
7 **Description of the parameters of the cases using to determine the best crossover type.** . . . 6
8 Determine the best population size . . . 6

List of Figures